

$\int_a^b f(x)dx$
 $\lim_{x \rightarrow 0} f(x)$
 $\frac{dy}{dx}$
 $ax + by \leq c$
 $\sqrt{x^2 + y^2}$

UNIT

2

DIFFERENTIATION

- Differentiation deals with the rate of change of a dependent variable with respect to one or more independent variables.
- In the function of the form $y = f(x)$ where $x \in \text{Dom } f$, x is called independent variable while y is called the dependent variable.

Derivative of a Function:

Let f be a real valued function continuous in the interval $(x, x_1) \subseteq D_f$ then

$\frac{f(x_1) - f(x)}{x_1 - x}$ is called average rate of change of the function.

If x_1 approaches to x then $\lim_{x_1 \rightarrow x} \frac{f(x_1) - f(x)}{x_1 - x}$ is called the instantaneous rate of change

of function with respect to x and is written as $f'(x)$ (read as "f prime of x")

Finding $f'(x)$ from definition of derivative:

$$\text{If } y = f(x) \tag{i}$$

$$\text{Step 1} \quad y + \delta y = f(x + \delta x) \tag{ii}$$

Subtracting equation (i) from equation (ii)

$$\text{Step 2} \quad \delta y = f(x + \delta x) - f(x)$$

$$\text{Step 3} \quad \frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

$$\text{Step 4} \quad \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

and

$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}$ is denoted by $\frac{dy}{dx}$, so

$$\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

Note:

The symbol $\frac{dy}{dx}$ is used for the derivative of y with respect to x and is not a quotient of dy and dx .

Name of Mathematician	Leibniz	Newton	Lagrange	Cauchy
Notation used for derivative	$\frac{dy}{dx}$ or $\frac{df}{dx}$	$f'(x)$	$f'(x)$	$Df(x)$

The Power Rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Case I: when $n \in \mathbb{Z}^+$

Proof:

$$\text{Let } y = x^n$$

$$y + \delta y = (x + \delta x)^n$$

$$\delta y = (x + \delta x)^n - y$$

$$\delta y = (x + \delta x)^n - x^n$$

Using the binomial theorem we have

$$\delta y = \left[x^n + nx^{n-1}\delta x + \frac{n(n-1)}{2!}x^{n-2}(\delta x)^2 + \dots + (\delta x)^n \right] - x^n$$

$$\delta y = \cancel{x^n} + nx^{n-1}\delta x + \frac{n(n-1)}{2!}x^{n-2}(\delta x)^2 + \dots + (\delta x)^n - \cancel{x^n}$$

$$\delta y = \delta x \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}\delta x + \dots + (\delta x)^{n-1} \right]$$

$$\frac{\delta y}{\delta x} = \frac{\cancel{\delta x}}{\cancel{\delta x}} \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}\delta x + \dots + (\delta x)^{n-1} \right]$$

Taking the limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}\delta x + \dots + (\delta x)^{n-1} \right]$$

$$\frac{dy}{dx} = nx^{n-1}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Case II: When $n \in \mathbb{Z}^-$

Let

$$n = -m$$

$$y = x^{-m}$$

$$y + \delta y = (x + \delta x)^{-m}$$

$$\delta y = (x + \delta x)^{-m} - y$$

$$= (x + \delta x)^{-m} - x^{-m}$$

$$= \left[x \left(1 + \frac{\delta x}{x} \right) \right]^{-m} - x^{-m}$$

$$= x^{-m} \left(1 + \frac{\delta x}{x} \right)^{-m} - x^{-m}$$

$$= x^{-m} \left[\left(1 + \frac{\delta x}{x} \right)^{-m} - 1 \right]$$

Using binomial series

$$\delta y = x^{-m} \left[1 - m \frac{\delta x}{x} + \frac{-m(-m-1)}{2!} \frac{(\delta x)^2}{x^2} + \dots - 1 \right]$$

$$= x^{-m} \left[\frac{-m\delta x}{x} + \frac{-m(-m-1)}{2!} \frac{(\delta x)^2}{x^2} + \dots \right]$$

$$\delta y = \delta x x^{-m} \left[\frac{-m}{x} + \frac{-m(-m-1)}{2!} \frac{\delta x}{x^2} + \dots \right]$$

$$\frac{\delta y}{\delta x} = \frac{\delta x}{\delta x} x^{-m} \left[\frac{-m}{x} + \frac{-m(-m-1)}{2!} \frac{\delta x}{x^2} + \dots \right]$$

Taking the limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \left(\frac{\delta y}{\delta x} \right) = \lim_{\delta x \rightarrow 0} x^{-m} \left[\frac{-m}{x} + \frac{-m(-m-1)}{2!} \frac{\delta x}{x^2} + \dots \right]$$

$$\frac{dy}{dx} = x^{-m} \left[\frac{-m}{x} + 0 + \dots \right]$$

$$\frac{d}{dx} (x^{-m}) = -mx^{-m-1}$$

Above rule also holds for $n \in \mathbb{Q} - \mathbb{Z}$

EXERCISE 2.1

Q.1 Find by definition, the derivatives w.r.t 'x' of the following functions defined as:

(i) $2x^2 + 1$

Solution:

Let $y = 2x^2 + 1$

Taking increments on both sides,

$$y + \delta y = 2(x + \delta x)^2 + 1$$

$$\delta y = 2(x^2 + \delta x^2 + 2x\delta x) + 1 - y$$

$$\delta y = \cancel{2x^2} + 2\delta x^2 + 4x\delta x + \cancel{1} - \cancel{2x^2} - \cancel{1}$$

$$\delta y = 2\delta x^2 + 4x\delta x$$

$$\delta y = \delta x(2\delta x + 4x)$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = 2\delta x + 4x$$

Taking limit when $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (4x + 2\delta x)$$

$$\boxed{\frac{dy}{dx} = 4x}$$

(ii) $2 - \sqrt{x}$

Solution:

Let $y = 2 - \sqrt{x}$

Taking increments on both sides,

$$y + \delta y = 2 - \sqrt{x + \delta x}$$

$$\delta y = 2 - \sqrt{x + \delta x} - y$$

$$\delta y = \cancel{2} - \sqrt{x + \delta x} - \cancel{2} + \sqrt{x}$$

$$\delta y = \sqrt{x} - \sqrt{x + \delta x}$$

$$\delta y = x^{\frac{1}{2}} - (x + \delta x)^{\frac{1}{2}}$$

$$\delta y = x^{\frac{1}{2}} - x^{\frac{1}{2}} \left(1 + \frac{\delta x}{x} \right)^{\frac{1}{2}}$$

$$\delta y = x^{\frac{1}{2}} \left[1 - \left(1 + \frac{\delta x}{x} \right)^{\frac{1}{2}} \right]$$

$$\delta y = x^{\frac{1}{2}} \left[1 - \left(1 + \frac{1}{2} \frac{\delta x}{x} + \frac{1}{2} \left(-\frac{1}{2} \right) \left(\frac{\delta x}{x} \right)^2 + \dots \right) \right]$$

$$\delta y = x^{\frac{1}{2}} \left[\cancel{1} - \cancel{1} - \frac{1}{2} \frac{\delta x}{x} + \frac{1}{8} \frac{\delta x^2}{x^2} + \dots \right]$$

$$\delta y = -x^{\frac{1}{2}} \delta x \left[\frac{1}{2x} - \frac{\delta x}{8x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -x^{\frac{1}{2}} \left[\frac{1}{2x} - \frac{\delta x}{8x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[-x^{\frac{1}{2}} \left(\frac{1}{2x} - \frac{\delta x}{8x^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = -\frac{1}{2\sqrt{x}}}$$

(iii) $\frac{1}{\sqrt{x}}$

Solution:

Let $y = \frac{1}{\sqrt{x}}$

Taking increments on both sides,

$$y + \delta y = \frac{1}{\sqrt{x + \delta x}}$$

$$\delta y = \frac{1}{\sqrt{x + \delta x}} - y$$

$$\delta y = \frac{1}{\sqrt{x + \delta x}} - \frac{1}{\sqrt{x}}$$

$$\delta y = (x + \delta x)^{-\frac{1}{2}} - x^{-\frac{1}{2}}$$

$$\delta y = x^{-\frac{1}{2}} \left(1 + \frac{\delta x}{x} \right)^{-\frac{1}{2}} - x^{-\frac{1}{2}}$$

$$= x^{\frac{-1}{2}} \left[\left(1 + \frac{\delta x}{x} \right)^{\frac{-1}{2}} - 1 \right]$$

$$= x^{\frac{-1}{2}} \left[\cancel{1} + \left(\frac{-1}{2} \right) \left(\frac{\delta x}{x} \right) + \frac{\left(\frac{-1}{2} \right) \left(\frac{-3}{2} \right)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$= \delta x \cdot x^{\frac{-1}{2}} \left[\frac{-1}{2x} + \frac{3\delta x}{8x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -x^{\frac{-1}{2}} \left[\frac{1}{2x} - \frac{3\delta x}{8x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$ both sides.

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[-x^{\frac{-1}{2}} \left(\frac{1}{2x} - \frac{3\delta x}{8x^2} + \dots \right) \right]$$

$$\frac{dy}{dx} = -x^{\frac{-1}{2}} \cdot \frac{1}{2x}$$

$$\boxed{\frac{dy}{dx} = \frac{-1}{2x^{\frac{3}{2}}}}$$

(iv) $\frac{1}{x^3}$

Solution:

Let $y = \frac{1}{x^3}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^{-3}$$

$$\delta y = (x + \delta x)^{-3} - y$$

$$\delta y = (x + \delta x)^{-3} - x^{-3}$$

$$\delta y = x^{-3} \left(1 + \frac{\delta x}{x} \right)^{-3} - x^{-3}$$

$$\delta y = x^{-3} \left[\left(1 + \frac{\delta x}{x} \right)^{-3} - 1 \right]$$

$$\delta y = x^{-3} \left[\cancel{1} + (-3) \left(\frac{\delta x}{x} \right) + \frac{(-3)(-4)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = -x^{-3} \delta x \left[\frac{3}{x} - \frac{6\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -x^{-3} \left[\frac{3}{x} - \frac{6\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[-x^{-3} \left(\frac{3}{x} - \frac{6\delta x}{x^2} + \dots \right) \right]$$

$$\frac{\delta y}{\delta x} = -x^{-3} \cdot \frac{3}{x}$$

$$\boxed{\frac{dy}{dx} = \frac{-3}{x^4}}$$

(v) $\frac{1}{x-a}$

Solution:

Let $y = \frac{1}{x-a}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x - a)^{-1}$$

$$\delta y = (x + \delta x - a)^{-1} - (x - a)^{-1}$$

$$\delta y = (x - a + \delta x)^{-1} - (x - a)^{-1}$$

$$\delta y = (x - a)^{-1} \left(1 + \frac{\delta x}{x - a} \right)^{-1} - (x - a)^{-1}$$

$$= (x - a)^{-1} \left[\left(1 + \frac{\delta x}{x - a} \right)^{-1} - 1 \right]$$

$$\delta y = (x - a)^{-1} \left[\cancel{1} + (-1) \left(\frac{\delta x}{x - a} \right) + \frac{(-1)(-2)}{2} \frac{\delta x^2}{(x - a)^2} + \dots \cancel{1} \right]$$

$$\delta y = -(x - a)^{-1} \left[\frac{\delta x}{x - a} - \frac{\delta x^2}{(x - a)^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -(x - a)^{-1} \left[\frac{1}{x - a} - \frac{\delta x}{(x - a)^2} + \dots \right]$$

Taking $\lim_{\delta x \rightarrow 0}$ both sides,

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[-\frac{1}{x-a} \left(\frac{1}{x-a} - \frac{\delta x}{(x-a)^2} + \dots \right) \right]$$

$$\frac{dy}{dx} = \frac{-1}{x-a} \left(\frac{1}{x-a} \right)$$

$$\boxed{\frac{dy}{dx} = -\frac{1}{(x-a)^2}}$$

(vi) $x(x-3)$

Solution:

Let $y = x(x-3)$

$$\Rightarrow y = x^2 - 3x$$

Taking increments both sides

$$y + \delta y = (x + \delta x)^2 - 3(x + \delta x)$$

$$\delta y = \cancel{x^2} + \delta x^2 + 2x\delta x - \cancel{3x} - 3\delta x - \cancel{x^2} + \cancel{3x}$$

$$\delta y = \delta x^2 + 2x\delta x - 3\delta x$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = \delta x + 2x - 3$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} [\delta x + 2x - 3]$$

$$\boxed{\frac{dy}{dx} = 2x - 3}$$

(vii) $\frac{2}{x^4}$

Solution:

Let $y = \frac{2}{x^4}$

Taking increments on both sides,

$$y + \delta y = 2(x + \delta x)^{-4}$$

$$\delta y = 2x^{-4} \left(1 + \frac{\delta x}{x} \right)^{-4} - 2x^{-4}$$

$$\delta y = 2x^{-4} \left[1 + (-4) \left(\frac{\delta x}{x} \right) + \frac{(-4)(-5)}{2} \frac{\delta x^2}{x^2} + \dots - 1 \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = 2x^{-4} \left[\cancel{1} - \frac{4}{x} + \frac{10\delta x}{x^2} + \dots \cancel{1} \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[2x^{-4} \left(-\frac{4}{x} + 10 \frac{\delta x}{x^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = \frac{-8}{x^5}}$$

(viii) $(x+4)^{\frac{1}{3}}$

Solution:

Let $y = (x+4)^{\frac{1}{3}}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x + 4)^{\frac{1}{3}}$$

$$\delta y = (x + 4 + \delta x)^{\frac{1}{3}} - (x + 4)^{\frac{1}{3}}$$

$$\delta y = (x + 4)^{\frac{1}{3}} \left[\left(1 + \frac{\delta x}{x+4} \right)^{\frac{1}{3}} - 1 \right]$$

$$\delta y = (x + 4)^{\frac{1}{3}} \left[\cancel{1} + \frac{1}{3} \frac{\delta x}{x+4} + \frac{\frac{1}{3} \left(\frac{-2}{3} \right)}{2} \frac{\delta x^2}{(x+4)^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x (x + 4)^{\frac{1}{3}} \left[\frac{1}{3(x+4)} - \frac{1}{9} \cdot \frac{\delta x}{(x+4)^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = (x + 4)^{\frac{1}{3}} \left[\frac{1}{3(x+4)} - \frac{\delta x}{9(x+4)^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (x + 4)^{\frac{1}{3}} \left[\frac{1}{3(x+4)} - \frac{1}{9} \cdot \frac{\delta x}{(x+4)^2} + \dots \right]$$

$$\frac{dy}{dx} = \frac{(x+4)^{\frac{1}{3}}}{3(x+4)}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{3(x+4)^{\frac{2}{3}}}}$$

(ix) $x^{\frac{3}{2}}$
Solution:

Let $y = x^{\frac{3}{2}}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^{\frac{3}{2}}$$

$$\delta y = x^{\frac{3}{2}} \left(1 + \frac{\delta x}{x} \right)^{\frac{3}{2}} - y$$

$$\delta y = x^{\frac{3}{2}} \left[\left(1 + \frac{\delta x}{x} \right)^{\frac{3}{2}} - 1 \right]$$

$$\delta y = x^{\frac{3}{2}} \left[\cancel{1} + \frac{3}{2} \cdot \frac{\delta x}{x} + \frac{3 \left(\frac{1}{2} \right)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x \cdot x^{\frac{3}{2}} \left[\frac{3}{2x} + \frac{3}{8} \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = x^{\frac{3}{2}} \left[\frac{3}{2x} + \frac{3\delta x}{8x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{3}{2} \cdot \frac{x^{\frac{3}{2}}}{x}$$

$$\boxed{\frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}}}$$

(x) $x^{\frac{5}{2}}$
Solution:

Let $y = x^{\frac{5}{2}}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^{\frac{5}{2}}$$

$$\delta y = x^{\frac{5}{2}} \left(1 + \frac{\delta x}{x} \right)^{\frac{5}{2}} - x^{\frac{5}{2}}$$

$$\delta y = x^{\frac{5}{2}} \left[\left(1 + \frac{\delta x}{x} \right)^{\frac{5}{2}} - 1 \right]$$

$$\delta y = x^{\frac{5}{2}} \left[\cancel{1} + \frac{5}{2} \left(\frac{\delta x}{x} \right) + \frac{5}{2} \left(\frac{3}{2} \right) \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x \cdot x^{\frac{5}{2}} \left[\frac{5}{2x} + \frac{15}{8} \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = x^{\frac{5}{2}} \left[\frac{5}{2x} + \frac{15}{8} \frac{\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} x^{\frac{5}{2}} \left[\frac{5}{2x} + \frac{15}{8} \frac{\delta x}{x^2} + \dots \right]$$

$$\frac{dy}{dx} = x^{\frac{5}{2}} \cdot \frac{5}{2x}$$

$$\boxed{\frac{dy}{dx} = \frac{5}{2} x^{\frac{3}{2}}}$$

(xi) x^m , $m \in \mathbb{Q}$

Solution:

Let $y = x^m$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^m$$

$$\delta y = (x + \delta x)^m - x^m$$

$$\delta y = x^m \left(1 + \frac{\delta x}{x} \right)^m - x^m$$

$$\delta y = x^m \left[\left(1 + \frac{\delta x}{x} \right)^m - 1 \right]$$

$$\delta y = x^m \left[\cancel{1} + m \frac{\delta x}{x} + \frac{m(m-1)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = x^m \delta x \left[\frac{m}{x} + \frac{m(m-1)}{2} \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = x^m \left[\frac{m}{x} + \frac{m(m-1)}{2} \frac{\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} x^m \left[\frac{m}{x} + \frac{m(m-1)}{2} \frac{\delta x}{x^2} + \dots \right]$$

$$\boxed{\frac{dy}{dx} = mx^{m-1}}$$

(xii) $\frac{1}{x^m}, m \in \mathbb{Q}$

Solution:

Let $y = \frac{1}{x^m}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^{-m}$$

$$\delta y = x^{-m} \left(1 + \frac{\delta x}{x} \right)^{-m}$$

$$\delta y = x^{-m} \left[\left(1 + \frac{\delta x}{x} \right)^{-m} - 1 \right]$$

$$\delta y = x^{-m} \left[\cancel{1} + (-m) \left(\frac{\delta x}{x} \right) + \frac{(-m)(-m-1)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x \cdot x^{-m} \left[\frac{-m}{x} + \frac{-m(-m-1)}{2} \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -x^{-m} \left[\frac{m}{x} + \frac{m(m+1)}{2} \frac{\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[-x^{-m} \left(\frac{m}{x} + \frac{m(m+1)}{2} \frac{\delta x}{x^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = \frac{-m}{x^{m+1}}}$$

(xiii) x^{40}

Solution:

Let $y = x^{40}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x)^{40}$$

$$\delta y = (x + \delta x)^{40} - x^{40}$$

$$\delta y = x^{40} \left[\left(1 + \frac{\delta x}{x} \right)^{40} - 1 \right]$$

$$\delta y = x^{40} \left[\cancel{1} + 40 \cdot \frac{\delta x}{x} + \frac{40(39)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x \cdot x^{40} \left[\frac{40}{x} + 780 \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = x^{40} \left[\frac{40}{x} + 780 \frac{\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[x^{40} \cdot \left(\frac{40}{x} + 780 \frac{\delta x}{x^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = 40x^{39}}$$

(xiv) x^{-100}

Solution:

Let $y = x^{-100}$

$$y + \delta y = (x + \delta x)^{-100}$$

$$\delta y = (x + \delta x)^{-100} - x^{-100}$$

$$\delta y = x^{-100} \left[\left(1 + \frac{\delta x}{x} \right)^{-100} - 1 \right]$$

$$\delta y = x^{-100} \left[\cancel{1} + (-100) \frac{\delta x}{x} + \frac{(-100)(-101)}{2} \frac{\delta x^2}{x^2} + \dots \cancel{1} \right]$$

$$\delta y = \delta x \cdot x^{-100} \left[\frac{-100}{x} + 780 \frac{\delta x}{x^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = x^{-100} \left[\frac{-100}{x} + 780 \frac{\delta x}{x^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[x^{-100} \cdot \left(\frac{-100}{x} - 5050 \frac{\delta x}{x^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = -100x^{-101}}$$

Q.2 Find $\frac{dy}{dx}$ from first principle if

(i) $\sqrt{x+2}$

Solution:

Let $y = \sqrt{x+2}$

$$\delta y = (x+2)^{\frac{1}{2}} \left[1 + \frac{1}{2} \frac{\delta x}{x+2} + \frac{\frac{1}{2} \left(-\frac{1}{2} \right)}{2} \frac{\delta x^2}{(x+2)^2} + \dots \right]$$

$$\delta y = (x+2)^{\frac{1}{2}} \cdot \delta x \left[\frac{1}{2(x+2)} - \frac{1}{8} \frac{\delta x}{(x+2)^2} + \dots \right]$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = (x+2)^{\frac{1}{2}} \left[\frac{1}{2(x+2)} - \frac{\delta x}{8(x+2)^2} + \dots \right] \text{ Taking limit as } \delta x \rightarrow 0$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[(x+2)^{\frac{1}{2}} \left(\frac{1}{2(x+2)} - \frac{\delta x}{8(x+2)^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = \frac{1}{2\sqrt{x+2}}}$$

(ii) $\frac{1}{\sqrt{x+a}}$

Solution:

Let $y = \frac{1}{\sqrt{x+a}}$

Taking increments on both sides,

$$y + \delta y = (x + \delta x + a)^{-\frac{1}{2}}$$

$$\delta y = (x + a + \delta x)^{-\frac{1}{2}} - (x + a)^{-\frac{1}{2}}$$

$$\delta y = (x + a)^{-\frac{1}{2}} \left[1 + \frac{\delta x}{x+a} \right]^{-\frac{1}{2}} - (x + a)^{-\frac{1}{2}}$$

Taking increments on both sides,

$$y + \delta y = (x + \delta x + 2)^{\frac{1}{2}}$$

$$\delta y = (x + 2 + \delta x)^{\frac{1}{2}} - (x + 2)^{\frac{1}{2}}$$

$$\delta y = (x + 2)^{\frac{1}{2}} \left[1 + \frac{\delta x}{x+2} \right]^{\frac{1}{2}} - (x + 2)^{\frac{1}{2}}$$

$$\delta y = (x + 2)^{\frac{1}{2}} \left[\left(1 + \frac{\delta x}{x+2} \right)^{\frac{1}{2}} - 1 \right]$$

$$\delta y = (x+a)^{-\frac{1}{2}} \left[\left(1 + \frac{\delta x}{x+a} \right)^{-\frac{1}{2}} - 1 \right]$$

$$\delta y = (x+a)^{-\frac{1}{2}} \left[1 + \frac{-1}{2} \left(\frac{\delta x}{x+a} \right) + \frac{\left(\frac{-1}{2} \right) \left(\frac{-3}{2} \right)}{2} \frac{\delta x^2}{(x+a)^2} + \dots \right]$$

$$\delta y = -(x+a)^{-\frac{1}{2}} \left[\frac{\delta x}{2(x+a)} - \frac{3}{8} \frac{\delta x^2}{(x+a)^2} + \dots \right]$$

Dividing both sides by ‘ δx ’

$$\frac{\delta y}{\delta x} = -\frac{1}{(x+a)^{\frac{1}{2}}} \left[\frac{1}{2(x+a)} - \frac{3}{8} \frac{\delta x}{(x+a)^2} + \dots \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[\frac{-1}{(x+a)^{\frac{1}{2}}} \left(\frac{1}{2(x+a)} - \frac{3}{8} \frac{\delta x}{(x+a)^2} + \dots \right) \right]$$

$$\boxed{\frac{dy}{dx} = \frac{-1}{2(x+a)^{\frac{3}{2}}}}$$

Differentiation of expressions of the types $(ax+b)^n$ and $\frac{1}{(ax+b)^n}$, $n=1,2,3,\dots$:

Let

$y = (ax+b)^n$, where n is a positive integer

$$y + \delta y = [a(x+\delta x) + b]^n = (ax + a\delta x + b)^n$$

$$\delta y = [(ax+b) + a\delta x]^n - (ax+b)^n$$

Using the binomial theorem

$$\delta y = (ax+b)^n + \binom{n}{1}(ax+b)^{n-1}(a\delta x) + \binom{n}{2}(ax+b)^{n-2}(a\delta x)^2 + \dots + (a\delta x)^n - (ax+b)^n$$

$$\delta y = \binom{n}{1}(ax+b)^{n-1}(a\delta x) + \binom{n}{2}(ax+b)^{n-2}a^2(\delta x)^2 + \dots + a^n(\delta x)^n$$

Dividing both sides by ‘ δx ’

$$\frac{\delta y}{\delta x} = \frac{\delta x}{\delta x} \left[\binom{n}{1}(ax+b)^{n-1}a + \binom{n}{2}(ax+b)^{n-2}a^2\delta x + \dots + a^n(\delta x)^{n-1} \right]$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left[\binom{n}{1} (ax+b)^{n-1} a + \binom{n}{2} (ax+b)^{n-2} a^2 \delta x + \dots + a^n (\delta x)^{n-1} \right]$$

$$\frac{dy}{dx} = n(ax+b)^{n-1} a + 0 + \dots + 0$$

$$\frac{d}{dx} (ax+b)^n = na(ax+b)^{n-1}$$